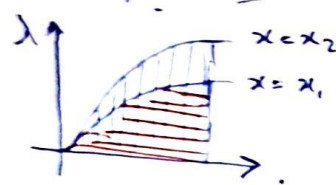
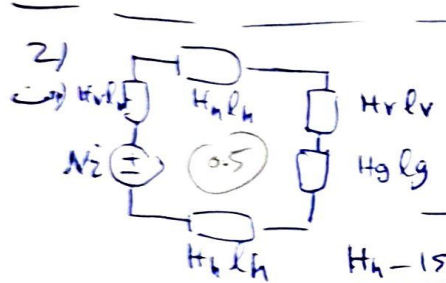


الف) با فرض ضل بودن مسطحه صافه، جگای جس اثرش در این $w_{Fe} = \frac{B_c^2}{2\mu_0}$ و در ناحیه حفری $w_{Fe} = \frac{B_c^2}{2\mu_0}$ با توجه به اینکه $\mu_0 \gg \mu_r \leftarrow w_{Fe} \gg w_{Fe}$ ، علیهم جم کم ناحیه حفری، یعنی اثرش آن بسیار است.



ب) با فرضی طاق فاصله حفری، لوکانش گامی و λ اثرش در این $w_{Fe} = \frac{B_c^2}{2\mu_0}$ تغییر یافته طاق $\leftarrow$ حفری ثابت $\leftarrow$ طبق شکل کانتور اثرش برابر (11111) $\leftarrow$ λ اثرش در این $w_{Fe} = \frac{B_c^2}{2\mu_0}$

$$\frac{P_{h,2}}{P_{h,1}} = \left(\frac{P_2}{P_1}\right)^{1-n} = 0.803, \quad \frac{P_{e,2}}{P_{e,1}} = 1 \rightarrow \frac{P_{e,2}}{P_{e,1}} = \frac{P_{e,1} + 0.803 P_{h,1}}{P_{e,1} + P_{h,1}} = 1 - 0.2 \frac{P_{h,1}}{P_{e,1}}$$



2) $B_g = \frac{\Phi_g}{A_g} = \frac{\alpha}{10 \times 10^{-4}} = 0.6 / 0.5 T \rightarrow H_g = \frac{B_g}{\mu_0} = \frac{0.6}{4\pi \times 10^{-7}} = 477.46 \times 10^3$
 $\Phi_g = \Phi_v = \Phi_n$ (No Fringing Flux) $\rightarrow B_v = 0.6 / 0.5 T$
 $H_v = 120 \times \frac{\alpha}{0.8} = 90 / 0.8 T$
 $\frac{H_n - 150}{B_h - 0.9} = \frac{900 - 150}{1.35 - 0.9} \rightarrow H_n = 650 / 316.67 A/m$
 $B_h = 1.2 / 1 T$
 $N_i = 2 H_v l_v + 2 H_n l_n + H_g l_g \rightarrow i = 0.807 / 0.645 A$
 $A_g \uparrow \rightarrow B_g \downarrow \rightarrow H_g \downarrow, H_v, H_n = cte \rightarrow i \downarrow$

3) $w_f = \int i d\lambda = \frac{0.04}{\lambda + 2} \left(\frac{2\lambda^{\beta+1}}{\beta+1} + \frac{\lambda^{\beta+\frac{3}{2}}}{\beta+\frac{3}{2}} \right) \Big|_{0.25}^{2.5} = 0.152 / 0.148 J$
 $i = 0.096 / 0.088 A \rightarrow w_f = i\lambda - w_f = 0.088 / 0.072 \rightarrow \frac{w_f}{w_p} = 1.73 / 2.06$
 $\rightarrow F_m = - \frac{\partial w_f}{\partial x} \Big|_{\lambda=cte} = \frac{0.04}{(\lambda+2)^2} \left(\frac{2\lambda^{\beta+1}}{\beta+1} + \frac{\lambda^{\beta+1.5}}{\beta+1.5} \right) = 0.071 / 0.069 N$

4) الف) $\Phi_1 \rightarrow R_A \rightarrow \Phi_{21} \rightarrow R_B \rightarrow \Phi_2 \rightarrow R_C \rightarrow \Phi_{21} \rightarrow R_A$
 $N_1 \rightarrow N_2$
 $R_A = R_B = \frac{6 \times 10^{-4}}{(1.03) \times 12 \times 10^{-4} \times \mu_0} = 3.86 \times 10^5 / 5.15 \times 10^5 (1/H)$
 $R_C = \frac{4 \times 10^{-4}}{1.03 \times 10 \times 10^{-4} \times \mu_0} = 3.09 \times 10^5 (1/H)$
 $L_{12} = \frac{N_2 \Phi_{21}}{i_1} \Big|_{i_2=0} \rightarrow R_{eq} = R_A + R_B \parallel R_C = 5.58 \times 10^5 / 7.08 \times 10^5 (1/H)$
 $\Phi_1 = \frac{N_1 i_1}{R_{eq}} \rightarrow \Phi_2 = \Phi_1 \frac{R_C}{R_B + R_C} \rightarrow \Phi_{21} = 0.445 \Phi_1 / 0.375 \Phi_1$
 $\rightarrow L_{12} = \frac{N_2 N_1 \times 0.445}{5.58 \times 10^5} = 7.97 / 5.30 mH$
 $R_A = 0 \rightarrow R_{eq} \downarrow, \Phi_2 = cte \rightarrow L_{12} \uparrow$

4) الف) $\Phi_1 \rightarrow R_A \rightarrow \Phi_{21} \rightarrow R_B \rightarrow \Phi_2 \rightarrow R_C \rightarrow \Phi_{21} \rightarrow R_A$
 $N_1 \rightarrow N_2$
 $R_A = R_B = \frac{6 \times 10^{-4}}{(1.03) \times 12 \times 10^{-4} \times \mu_0} = 3.86 \times 10^5 / 5.15 \times 10^5 (1/H)$
 $R_C = \frac{4 \times 10^{-4}}{1.03 \times 10 \times 10^{-4} \times \mu_0} = 3.09 \times 10^5 (1/H)$
 $L_{12} = \frac{N_2 \Phi_{21}}{i_1} \Big|_{i_2=0} \rightarrow R_{eq} = R_A + R_B \parallel R_C = 5.58 \times 10^5 / 7.08 \times 10^5 (1/H)$
 $\Phi_1 = \frac{N_1 i_1}{R_{eq}} \rightarrow \Phi_2 = \Phi_1 \frac{R_C}{R_B + R_C} \rightarrow \Phi_{21} = 0.445 \Phi_1 / 0.375 \Phi_1$
 $\rightarrow L_{12} = \frac{N_2 N_1 \times 0.445}{5.58 \times 10^5} = 7.97 / 5.30 mH$
 $R_A = 0 \rightarrow R_{eq} \downarrow, \Phi_2 = cte \rightarrow L_{12} \uparrow$